

## Pancreas Segmentation (512X512\_256x256\_128x128)

(Framework=PyTorch, Treshold=0.40, trainig size=0.8, vaidation size=0.1, test\_size=0.1)

(Train image=**938**, Test image=**118**, validation image=**117**)

### (Model=DeepLabV3+)

image size 512x512 (Model=DeepLabV3Plus)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.6081        | 0.6679         | 0.6761         |
| Precision              | 0.7339        | 0.7919         | 0.7650         |
| Recall                 | 0.7640        | 0.7897         | 0.8460         |
| Specificity            | 0.9942        | 0.9968         | 0.9954         |
| F1 Skore               | 0.7271        | 0.7763         | 0.7891         |
| Running Time (seconds) | <b>587.47</b> | <b>2365.56</b> | <b>5862.49</b> |

image size 256x256 (Model=DeepLabV3Plus)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.4934        | 0.5342         | 0.6183         |
| Precision              | 0.7272        | 0.6915         | 0.7142         |
| Recall                 | 0.5561        | 0.6671         | 0.8079         |
| Specificity            | 0.9968        | 0.9940         | 0.9939         |
| F1 Skore               | 0.6081        | 0.6546         | 0.7427         |
| Running Time (seconds) | <b>410.06</b> | <b>1175.77</b> | <b>1921.20</b> |

image size 128x128 (Model=DeepLabV3Plus)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.4730        | 0.5250         | 0.5532         |
| Precision              | 0.5691        | 0.6131         | 0.6453         |
| Recall                 | 0.6985        | 0.7522         | 0.7775         |
| Specificity            | 0.9868        | 0.9910         | 0.9907         |
| F1 Skore               | 0.5962        | 0.6515         | 0.6888         |
| Running Time (seconds) | <b>468.61</b> | <b>1333.59</b> | <b>1866.52</b> |

**(Model : U-Net++)**

image size 512x512 (Model: U-NetPlusPlus)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.5498        | 0.6546         | 0.6675         |
| Precision              | 0.5993        | 0.8048         | 0.8235         |
| Recall                 | 0.8208        | 0.7779         | 0.7654         |
| Specificity            | 0.9882        | 0.9958         | 0.9964         |
| F1 Skore               | 0.6745        | 0.7601         | 0.7765         |
| Running Time (seconds) | <b>732.72</b> | <b>3076.99</b> | <b>6451.02</b> |

image size 256x256 (Model: U-NetPlusPlus)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.6069        | 0.6304         | 0.6324         |
| Precision              | 0.7336        | 0.7730         | 0.7531         |
| Recall                 | 0.7639        | 0.7610         | 0.7786         |
| Specificity            | 0.9944        | 0.9958         | 0.9950         |
| F1 Skore               | 0.7281        | 0.7523         | 0.7532         |
| Running Time (seconds) | <b>607.93</b> | <b>1581.08</b> | <b>2758.55</b> |

image size 128x128 (Model: U-NetPlusPlus)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.5384        | 0.5518         | 0.5736         |
| Precision              | 0.6464        | 0.6884         | 0.7392         |
| Recall                 | 0.7468        | 0.7184         | 0.7004         |
| Specificity            | 0.9925        | 0.9937         | 0.9960         |
| F1 Skore               | 0.6628        | 0.6811         | 0.6934         |
| Running Time (seconds) | <b>376.13</b> | <b>1075.29</b> | <b>2677.30</b> |

**(Model : Segformer)**

**image size 512x512 (Model: Segformer)**

| Number of Epochs       | 10      | 30      | 50      |
|------------------------|---------|---------|---------|
| IoU                    | 0.5516  | 0.6398  | 0.6831  |
| Precision              | 0.6508  | 0.7183  | 0.7725  |
| Recall                 | 0.7749  | 0.8104  | 0.8559  |
| Specificity            | 0.9895  | 0.9940  | 0.9942  |
| F1 Skore               | 0.6813  | 0.7473  | 0.7997  |
| Running Time (seconds) | 1080.54 | 3106.61 | 5849.35 |

**image size 256x256 (Model: Segformer)**

| Number of Epochs       | 10     | 30     | 50      |
|------------------------|--------|--------|---------|
| IoU                    | 0.5278 | 0.5494 | 0.5901  |
| Precision              | 0.6193 | 0.7118 | 0.7297  |
| Recall                 | 0.7317 | 0.6824 | 0.7451  |
| Specificity            | 0.9903 | 0.9954 | 0.9937  |
| F1 Skore               | 0.6504 | 0.6723 | 0.7140  |
| Running Time (seconds) | 403.63 | 948.24 | 1958.42 |

**image size 128x128 (Model: Segformer)**

| Number of Epochs       | 10     | 30     | 50 (46 e.s.) |
|------------------------|--------|--------|--------------|
| IoU                    | 0.4942 | 0.5122 | 0.5797       |
| Precision              | 0.6805 | 0.6922 | 0.6828       |
| Recall                 | 0.6200 | 0.6293 | 0.7783       |
| Specificity            | 0.9941 | 0.9956 | 0.9938       |
| F1 Skore               | 0.6104 | 0.6293 | 0.7035       |
| Running Time (seconds) | 370.31 | 897.63 | 1736.12      |